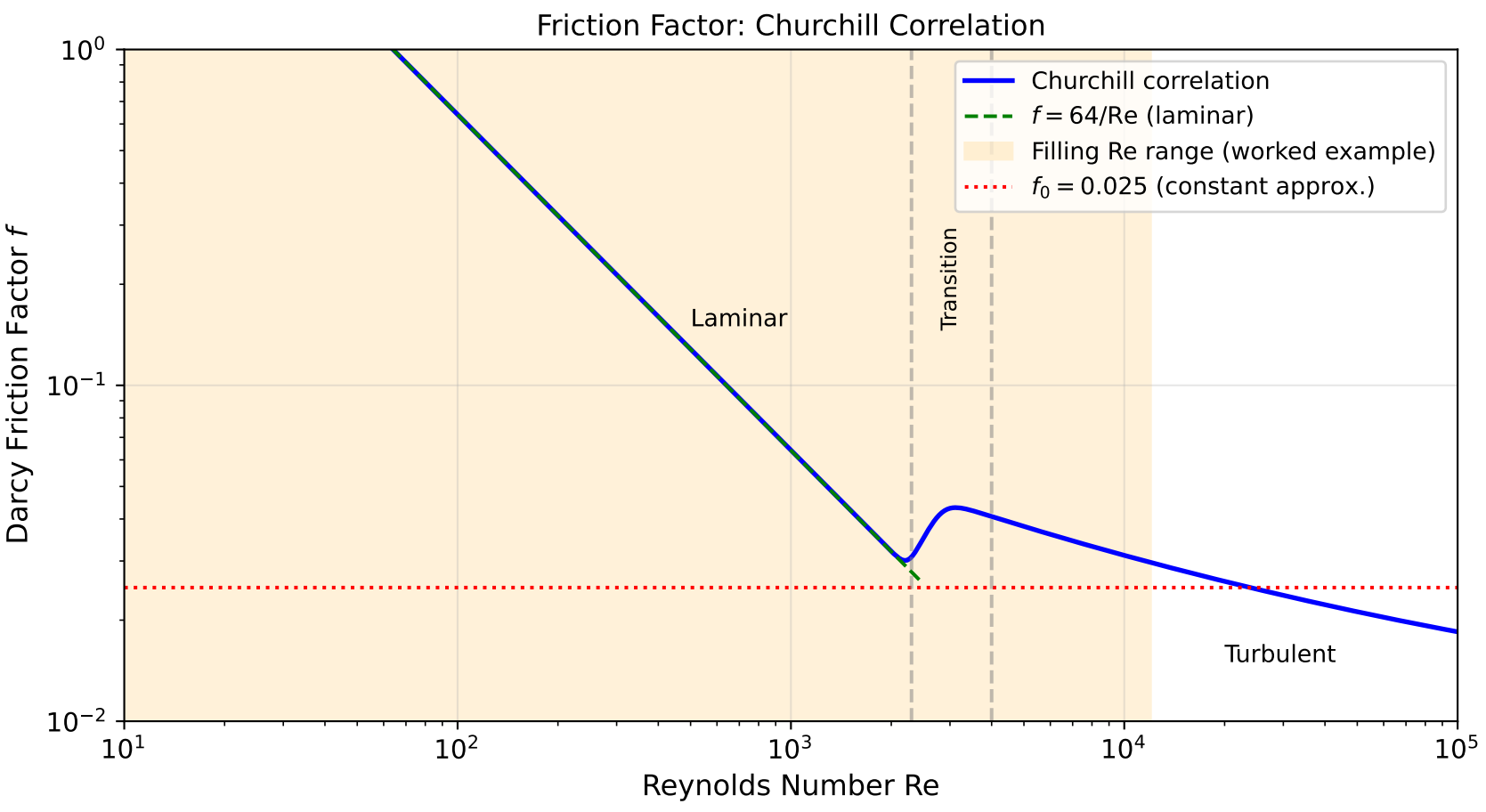
Friction Factor: Churchill Correlation
Churchill correlation
$f = 64/\mathrm{Re}$ (laminar)
Filling Re range (worked example)
$f_0 = 0.025$ (constant approx.)
Laminar
Transition
Turbulent
Darcy Friction Factor f
Reynolds Number Re
10^0
10^{-1}
10^{-2}
10^1
10^2
10^3
10^4
10^5